

Introduction to Big Data Analytics

Week 1 Lab — Python Foundations

Part 1 — Variables & Data Types

1.1

```
my_name = "Hakizimana Honore Patient"
my_age = 22
my_gpa = 15
am_i_present = True

print(my_name, type(my_name))
print(my_age, type(my_age))
print(my_gpa, type(my_gpa))
print(am_i_present, type(am_i_present))
```

1.2 — Type conversion

```
age_text = "21"
age = int(age_text)
print(age + 1)
```

Answer: 22

1.3 — Think about it

```
print("3" + "4")
```

Answer: 34. Explanation: "3" and "4" are strings, so + joins them instead of performing numerical addition.

Part 2 — Strings

2.1 — f-strings

```
print(f"I am {my_name}, {my_age} years old, studying at AUCA.")
```

2.2 — String tools

```
course = "Big Data Analytics"

print(course.upper())
print(len(course))
```

```
print(course[0:3])
```

Answers: BIG DATA ANALYTICS; 18 characters; Big

Part 3 — Lists

3.1

```
scores = [72, 85, 91, 64, 78]

print("First score:", scores[0])
print("Last score:", scores[-1])
print("Number of scores:", len(scores))
print("Average:", sum(scores) / len(scores))
print("Highest score:", max(scores))
print("Lowest score:", min(scores))
```

Answers: First = 72; Last = 78; Number = 5; Average = 78; Highest = 91; Lowest = 64.

3.2 — Lists change

```
scores.append(88)
scores[3] = 66

print(scores)
print("New average:", sum(scores) / len(scores))
```

Answer: Updated list = [72, 85, 91, 66, 78, 88]; New average = 80.0.

Part 4 — Dictionaries

4.1

```
me = {
    "name": "Hakizimana Honore Patient",
    "age": 22,
    "program": "Law",
    "district": "Muhanga"
}

print(me["program"])
```

Answer: Law

4.2 — Update and extend

```
me["score"] = 85
me["age"] = me["age"] + 1
```

```
print(me.keys())
print(me)
```

Answer: The age becomes 22.

Part 5 — Class Dataset Analysis

5.1 — Describe the class

```
print(f"Number of students: {len(students)}")
print(f"Class average score: {round(sum(all_scores) / len(all_scores), 1)}")
print(f"Highest score: {max(all_scores)}")
print(f"Lowest score: {min(all_scores)}")
print(f"Range: {max(all_scores) - min(all_scores)}")
```

Answers: Number = 20; Class average = 76.7; Highest = 94; Lowest = 55; Range = 39.

5.2 — Inspect individual records

```
print(students[0])
print(students[-1])

s = students[6]
print(f"{s['name']} from {s['district']} studies {s['program']} and scored {s['score']}")
```

Answers: First = Aline Uwase; Last = Samuel Rukundo; Position 6 = Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.

5.3 — Count by category

```
print("Gasabo:", districts.count("Gasabo"))
print("Kicukiro:", districts.count("Kicukiro"))
print("Nyarugenge:", districts.count("Nyarugenge"))
```

Answers: Gasabo = 5; Kicukiro = 4; Nyarugenge = 3.

5.4 — Top student

```
best_score = max(all_scores)
position = all_scores.index(best_score)
top_name = names[position]

print(f"The top student is {top_name} with a score of {best_score}.")
```

Answer: The top student is Sandrine Umutoni with a score of 94.

Bonus — Lowest student

```
lowest_score = min(all_scores)
position = all_scores.index(lowest_score)
lowest_name = names[position]

print(f"The lowest student is {lowest_name} with a score of {lowest_score}.")
print(f"Attendance: {students[position]['attendance']}%")
```

Answer: The lowest student is David Hakizimana with a score of 55. Attendance = 72%.

Part 6 — Reflection

1. Is this 20-student dataset Big Data?

Answer: No. This dataset is not Big Data because it contains only 20 students and a small amount of information. It clearly does not demonstrate the Volume and Velocity characteristics of Big Data.

2. One thing that surprised or confused you

Answer: I was surprised that adding two numbers stored as strings produces 34 instead of 7. This helped me understand the importance of data types and type conversion in Python.